



Let the mind manage the body
Que l'esprit gère le corps

SCIENCE

(Subject code No. P141/3)

Index Number:

MAURITIUS EXAMINATIONS SYNDICATE

Primary School Achievement Certificate Assessment

2021-2022

Time: 1 hour 45 Minutes

Total Marks: 100

INSTRUCTIONS TO CANDIDATES

1. Check that this assessment booklet contains **7** questions printed on **19** pages numbered 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 and 20.
2. Write your Index Number on the assessment booklet in the space provided above.
3. You should not use red, green or black ink in answering questions.
4. Write all your answers clearly in the assessment booklet.
5. Attempt all questions.

Question	Marking		Revision		Control	
	Marks	Sig	Marks	Sig	Marks	Sig
1						
2						
3						
4						
5						
6						
7						
Total						
Sig (HoG)						

Question 1 (10 marks)

Circle the correct answer. Each item carries one mark.

(a) How many teeth are there in a set of permanent teeth?

A 10

B 16

C 20

D 32

(b) Which one of the following materials is impermeable?

A Cotton

B Nylon

C Silk

D Wool

(c) Which one of the following is an aquatic plant?

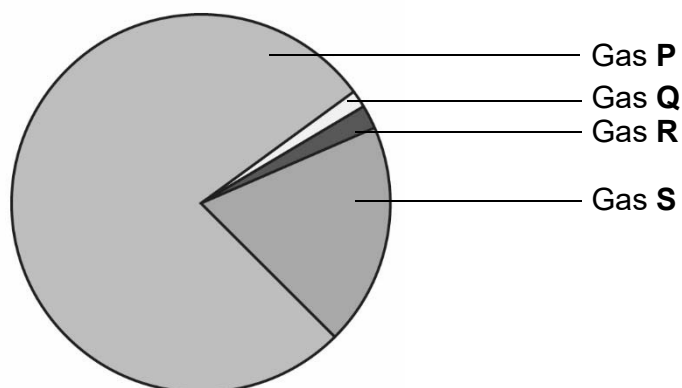
A Fern

B Lotus

C Moss

D Mugut

(d) The pie chart below represents the proportion of different gases in air.



Which part of the pie chart represents oxygen gas?

- A Gas P
- B Gas Q
- C Gas R
- D Gas S

(e) **Diagram 1** shows a hibiscus.

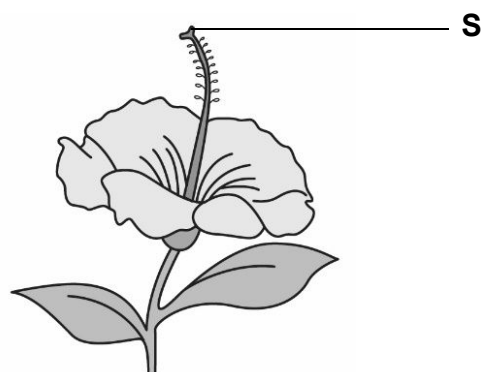


Diagram 1: A hibiscus

What is the function of Part **S**?

- A It produces pollen.
- B It manufactures food.
- C It attracts insects.
- D It receives pollen.

- (f) **Diagram 2** shows the water cycle.

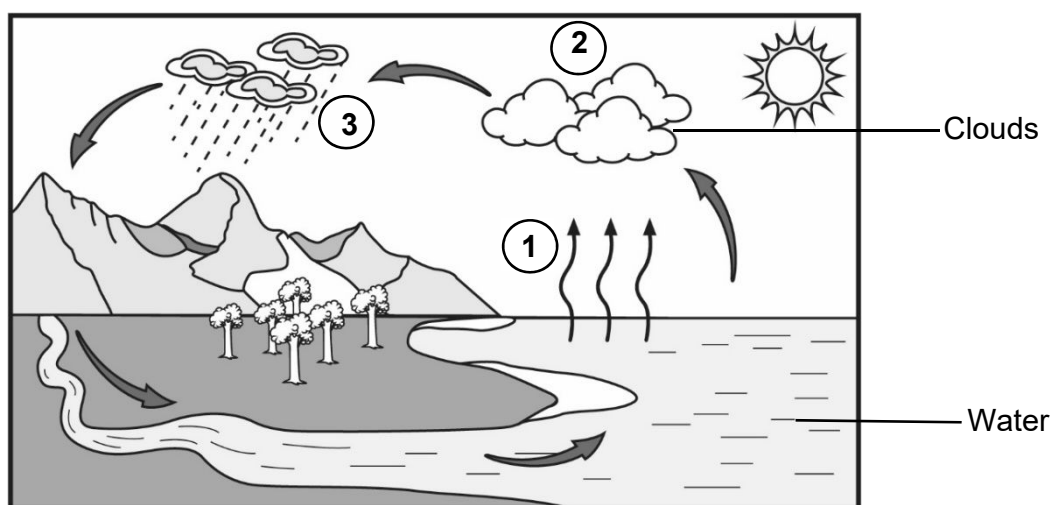


Diagram 2: The water cycle

What is the natural process occurring at (3)?

- A** Condensation
 - B** Evaporation
 - C** Melting
 - D** Precipitation
- (g) Which one of the following movements is completed in a year?
- A** One complete rotation of the Moon around the Earth
 - B** One complete rotation of the Earth on its axis
 - C** One complete rotation of the Earth around the Sun
 - D** One complete rotation of the Moon on its axis

- (h) **Diagram 3** shows solar panels on a house.

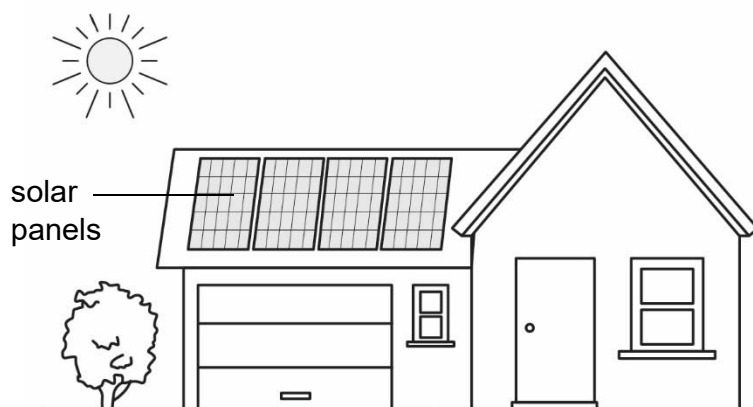


Diagram 3: Solar panels

Which **form** of energy is used by solar panels?

- A** Chemical energy
 - B** Heat energy
 - C** Light energy
 - D** Movement energy
- (i) Which one of the following birds found in Mauritius is an **exotic** bird?
- A** Echo Parakeet
 - B** Pink Pigeon
 - C** Solitaire
 - D** Sparrow
- (j) Ahmad places a cockroach in a closed jar.
What will happen to the amount of oxygen and carbon dioxide in the jar after some time?

	Amount of oxygen	Amount of carbon dioxide
A	decrease	decrease
B	increase	increase
C	increase	decrease
D	decrease	increase

QUESTION 2 (11 marks)

(a) **Diagram 4** shows a tomato plant.

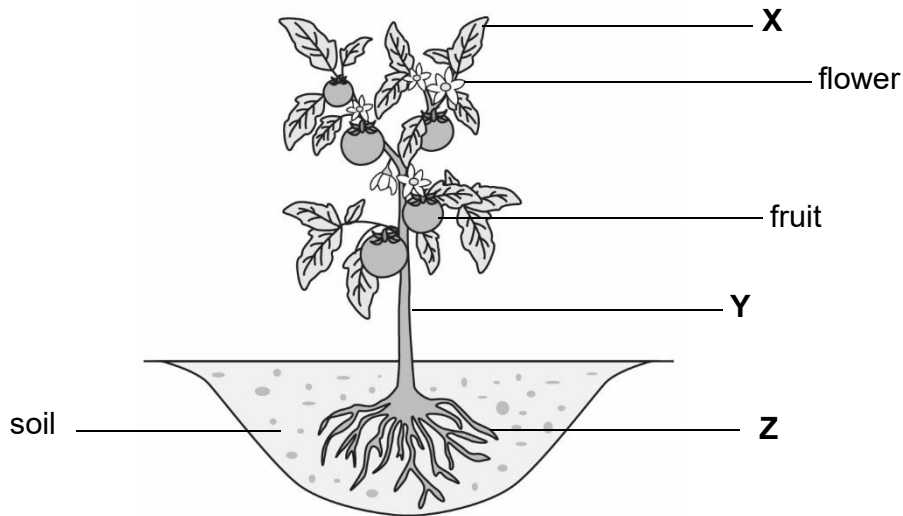


Diagram 4: A tomato plant

(i) Name parts **X** and **Y** of the plant.

X:

Y: [2]

(ii) Which part of the tomato plant holds the branches?

..... [1]

(iii) Give one function of Part **Z**.

..... [1]

(b) (i) What is pollination?

.....
..... [1]

(ii) Name an insect which is responsible for pollination.

..... [1]

(iii) Give two functions of the fruit.

1.

2. [2]

(c) (i) When a seed germinates, which part of the seedling grows first?

..... [1]

(ii) Give two conditions necessary for seeds to germinate.

1.

2. [2]

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QUESTION 3 (19 marks)

(a) Match each source of energy to the corresponding main form of energy it produces.

Source of energy

Form of energy produced

Cooking gas ●

Wind ●

Doorbell ●

● Movement

● Sound

● Chemical

● Heat

[3]

- (b) Rajen sets up an open circuit as shown in **Diagram 5**.

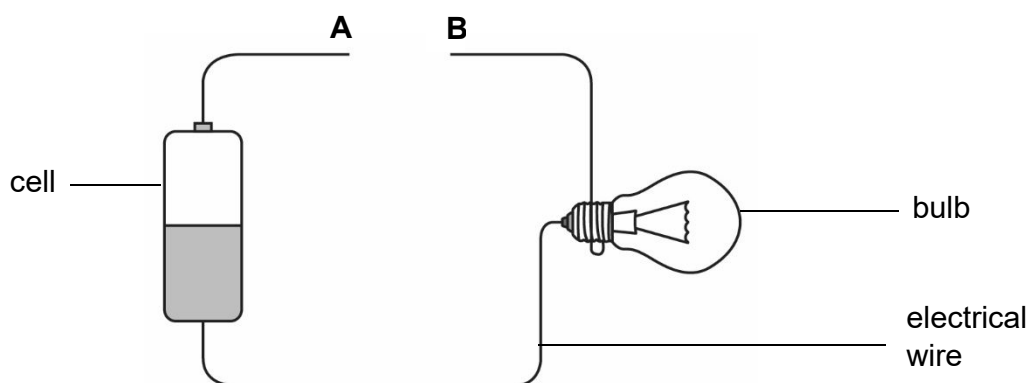


Diagram 5: An open circuit

- (i) What is an open circuit?
 [1]
- (ii) On **Diagram 5**, write plus (+) and minus (-) on the cell. [1]
- (c) Rajen places different objects, one at a time, between points **A** and **B**. He records his observations in **Table 1**.

- (i) Put a tick (✓) in the appropriate column in **Table 1** to indicate which object would cause the bulb to light up or not. An example is given.

Table 1

Name of object	Bulb lights up	Bulb does not light up
Plastic spoon		✓
Iron wire		
Dry piece of wood		
Aluminium coin		
Metal nail		

[4]

- (ii) Why do some of the objects in **Table 1** cause the bulb to light up?
 [1]

(iii) What is the function of the electrical wire in the circuit?

..... [1]

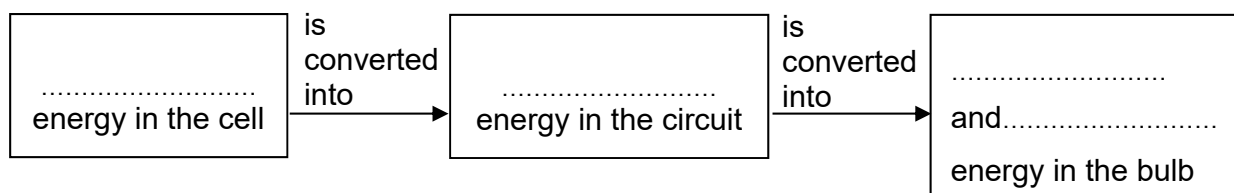
(iv) Why are electrical wires usually covered with rubber?

..... [1]

(v) State one precaution that should be taken when using electrical appliances.

..... [1]

(d) Fill in the boxes below to show the energy transformation that takes place when the bulb in the circuit lights up.



[4]

(e) Give two ways of saving energy.

1.

2.

[2]

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HG
QC
CE

QUESTION 4 (14 marks)

- (a) (i) Give one importance of a habitat to an animal.

..... [1]

- (ii) Complete **Table 2** by naming an animal which lives in the given habitat.

Table 2

Habitat	Animal
In the forest	
In the soil	

[2]

- (iii) Name one animal which lives both in water and on land.

..... [1]

- (iv) Fish live in water.

Name one part of the body of a fish that helps it to move about in water.

..... [1]

- (v) A hot desert is a habitat where very little rain falls.

Name a plant that grows well in hot deserts.

..... [1]

- (vi) Give two ways in which the plant you mentioned in part (a)(v) is adapted to live in its habitat.

1.

2. [2]

- (b) **Diagram 6** shows an ecosystem.

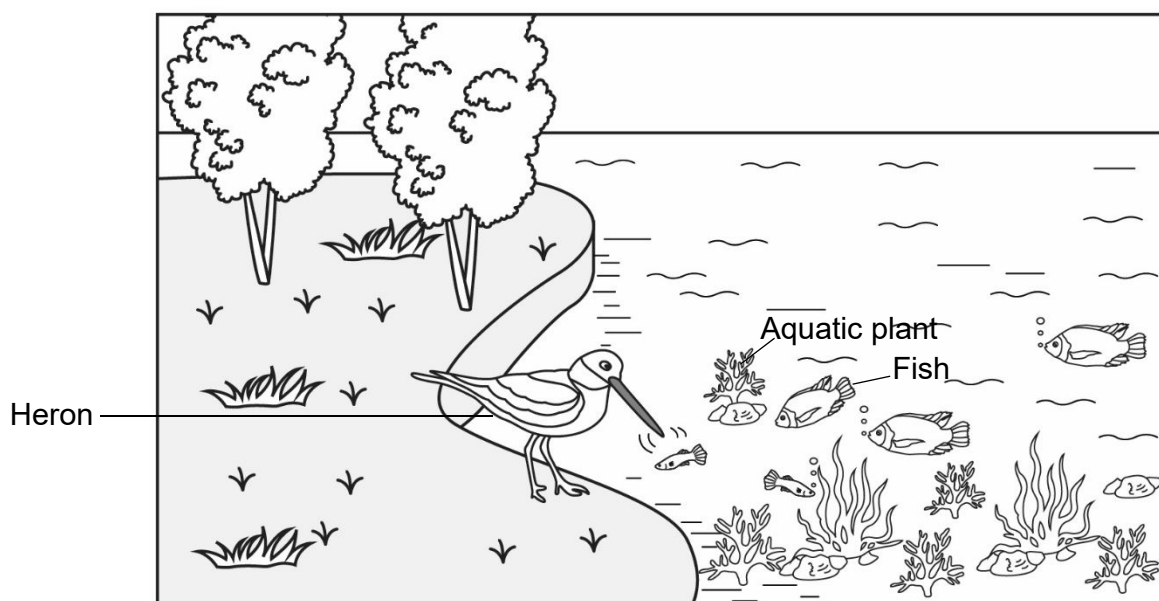
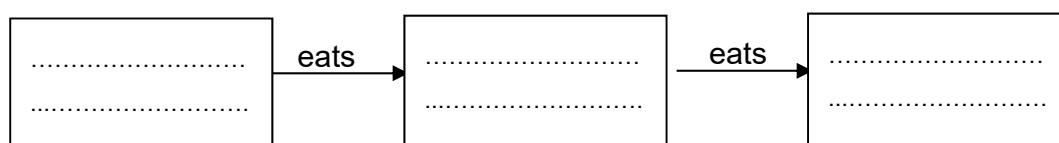


Diagram 6: An ecosystem

- (i) Fill in the boxes below to show how the living things in the ecosystem depend on each other.



[3]

- (ii) Name the process by which aquatic plants make their food.

.....

[1]

- (iii) Give two conditions necessary for the process you mentioned in part (b)(ii) to take place.

1.

2.

[2]

M
HG
QC
CE

QUESTION 5 (15 marks)

- (a) (i) One characteristic of living things is that they feed themselves.
Give another characteristic of living things.

..... [1]

- (ii) **Diagram 7** shows a tiger and a goat.

Choose the correct word from the list below and write down the food eaten by each animal in the spaces provided.

grains

flesh

grass

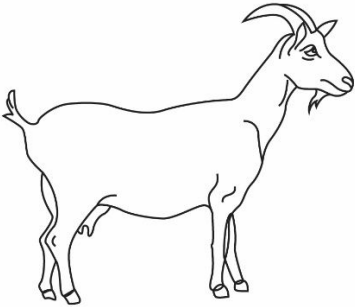
A tiger	A goat
	
Food it eats:	Food it eats:

Diagram 7: A tiger and a goat

- (iii) Observe the tiger.

[2]

1. What is the name given to its long pointy teeth?

..... [1]

2. State the function of these teeth.

..... [1]

- (iv) The monkey is an omnivorous animal.
What is an omnivorous animal?

..... [1]

(b) (i) What form of energy is stored in food?

.....

[1]

(ii) What do animals use energy for?

.....

[1]

(c) To stay in good health, we need to eat balanced meals.

(i) Name the three food groups that form a balanced meal.

1.

2.

3.

[3]

(ii) **Diagram 8** shows an incomplete meal for a **vegetarian**.

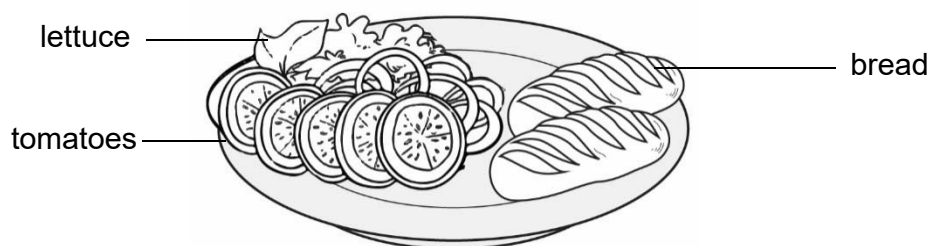


Diagram 8: Incomplete vegetarian meal

Which food group is missing from the meal?

.....

[1]

(iii) Which food item could be added to the meal in **Diagram 8** to make it a balanced vegetarian meal? Tick (✓) the correct box.

☐

fish

☐

carrots

☐

lentils

[1]

(iv) Apart from eating balanced meals, give two other ways of staying in good health.

1.

2. [2]

M
HG
QC
CE

QUESTION 6 (19 marks)

- (a) Materials can be classified as natural or man-made.
They may be obtained from different sources.

- (i) Cotton is obtained from a plant.
Give another example of a material that is obtained from a plant.

..... [1]

- (ii) Give two examples of materials obtained from animals.

1.

2. [2]

- (iii) Name a material that rusts very easily.

..... [1]

- (iv) State two conditions necessary for rusting to occur.

1.

2. [2]

- (b) Glass is used to make window panes.

- (i) Name the raw material used to make glass.

..... [1]

- (ii) Give one property of glass which makes it suitable to make window panes.

..... [1]

- (iii) Glass is a man-made material.
Give two other examples of man-made materials.

1.

2. [2]

(c) **Diagram 9** shows Ah Fat's guitar.

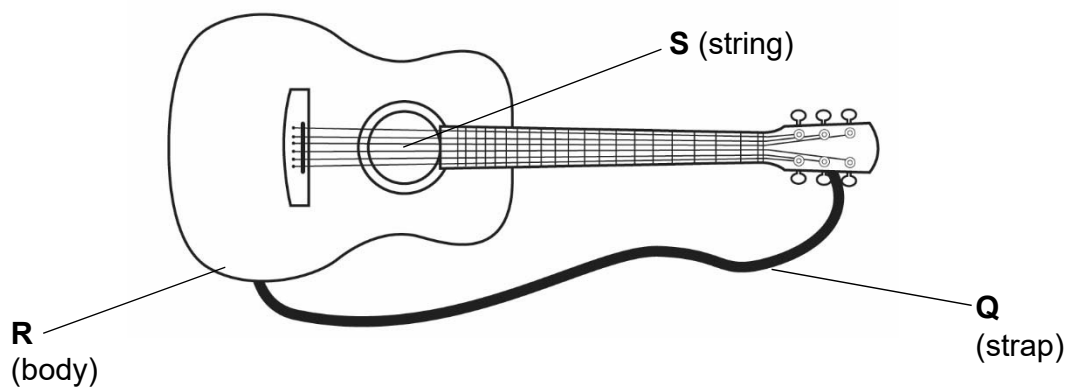


Diagram 9: Ah Fat's guitar

Suggest suitable materials that are commonly used to make parts **Q**, **R** and **S** of a guitar.

Q:

R:

S:

[3]

M
HG
QC
CE

- (d) (i) Name a gas needed for burning to take place.

.....

[1]

- (ii) An experiment to investigate the burning of candles is carried out. Three similar candles are lighted and covered using glass jars of different heights as shown in **Diagram 10**.

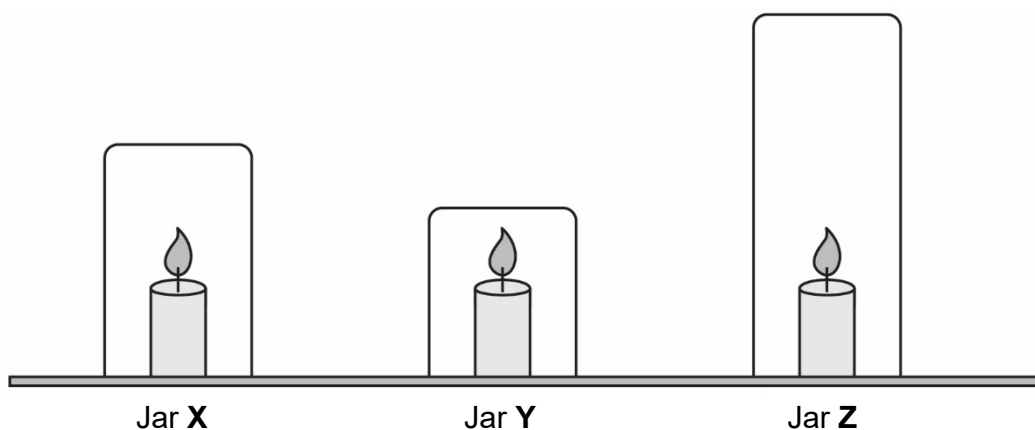


Diagram 10: Experiment to investigate the burning of candles

The time taken for the candles to extinguish is recorded in **Table 4**.

Table 4

Jar	Time
Y	3 seconds
Z	6 seconds

Suggest the time taken by the candle in Jar **X** to extinguish.

Tick (✓) the correct box.

☐

2 seconds

☐

5 seconds

☐

8 seconds

[1]

- (iii) Give a reason for your answer to part (d)(ii).

.....

.....

[1]

(e) **Diagram 11** shows the oxygen-carbon dioxide cycle.

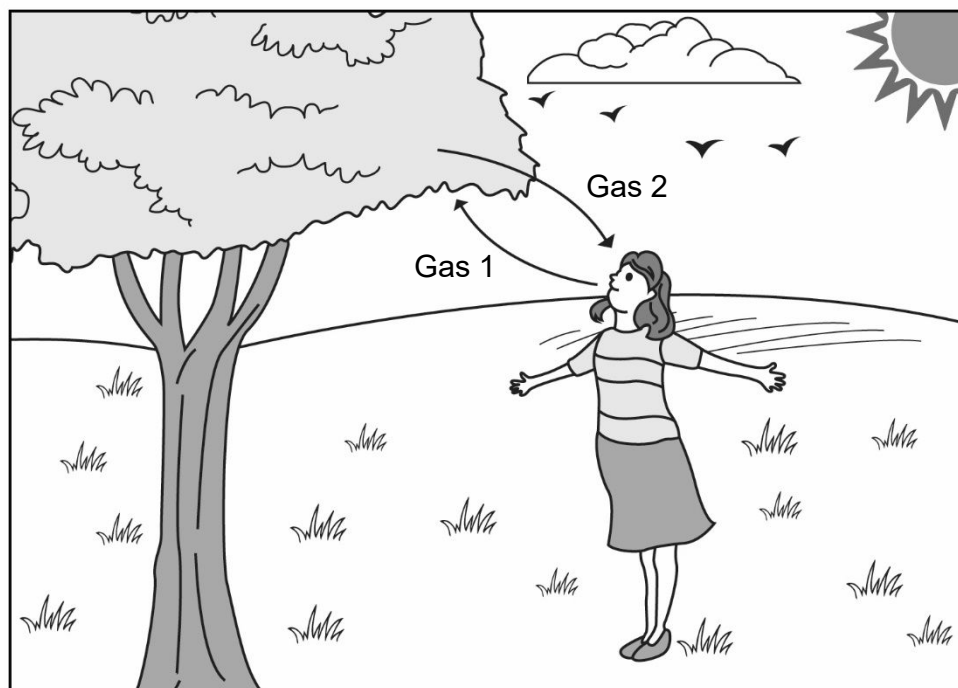


Diagram 11: oxygen-carbon dioxide cycle

(i) Identify Gas 1 and Gas 2.

Gas 1: Gas 2: [1]

(ii) Give two ways how the oxygen-carbon dioxide cycle is important.

1.
.....
2.
.....

[2]

M
HG
QC
CE

QUESTION 7 (12 marks)

- (a) **Diagram 12** shows a thermometer.

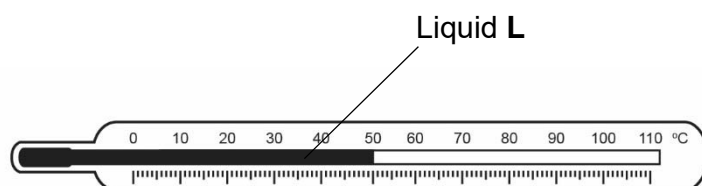


Diagram 12: A thermometer

Name Liquid **L**.

.....

[1]

- (b) **Diagram 13** shows two glasses of water, Glass **A** and Glass **B**. The temperature of the water in each glass is measured using the thermometer in turn.

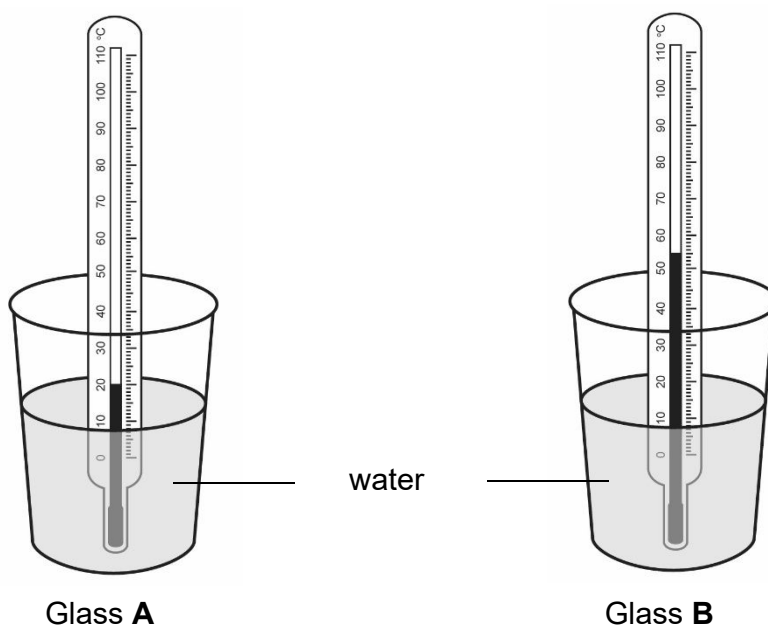


Diagram 13: Glass A and Glass B

- (i) In the table below, write down the temperatures shown in **Diagram 13**.

Glass	Temperature
A	°C
B	°C

[2]

- (ii) In which glass is the water warmer?

.....

[1]

- (iii) Some ice cubes are placed in Glass **B**.
What will happen to Liquid **L** in the thermometer?

.....

[1]

- (c) A bottle of cold drink is taken from the refrigerator.
After a few minutes, small droplets of water are seen on the outer surface of the bottle as shown in **Diagram 14**.

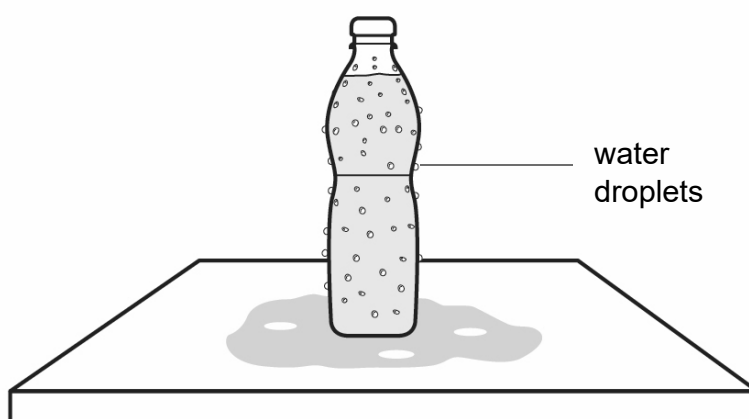


Diagram 14: Droplets of water on a bottle

- (i) Name the process by which these water droplets are formed.

.....

[1]

- (ii) Explain how the process mentioned in (c)(i) occurs.

.....

.....

[1]

- (d) To protect the environment, it is important to dispose of waste in appropriate ways.

- (i) Give one example of a waste from plants.

.....

[1]

- (ii) When waste from plants is left for some time, it becomes compost.
Give one use of compost.

..... [1]

- (iii) What type of pollution is caused by the burning of waste?

..... [1]

- (iv) Give one way how the dumping of plastic waste in rivers can be harmful.

.....
..... [1]

- (v) Give one way of disposing plastic waste that is safe for the environment.

..... [1]

M
HG
QC
CE

END OF PAPER